

Jadon Lai

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jadonlai.github.io/github-portfolio

EDUCATION

California Polytechnic State University, San Luis Obispo, CA

2021 - 2024

- **Major:** Bachelor's of Computer Science (AI and Machine Learning), open to obtaining a Master's
- **GPA:** 3.5 | **Dean's List:** 4 quarters | **President's List:** 2023-2024

SKILLS

- **AI:** Reinforcement Learning, Computational Neuroscience, Neural Networks, LLMs, Transformers, Tensorflow, PyTorch, SciKit-Learn, CUDA
- **Frontend:** UI/UX design, React, TailwindCSS, Web/Mobile Development
- **Computer Languages:** Python, TypeScript/JavaScript, C, HTML/CSS

EXPERIENCE

AI/ML Engineer | MyBackHub

Jan. 2026 - Present

- Designing, training, and deploying models trained on 3M+ patient records to diagnose patients with back problems
- Building scalable data pipelines and feature engineering workflows to clean, transform, and analyze the large-scale healthcare dataset for predictive modeling

Founder | Evros

Jun. 2024 - Present

- Solo-developing a cross-country management app to improve coach and fan interaction with student-athletes
- Full-stack application using JavaScript/React Native and Appwrite
- Key features include a custom attendance-taking page, synchronized race stopwatch, and team schedule

Sports Coach | SLO High School

Apr. 2023 - Present

- Coaching cross country and track and field athletes at SLO High School
- Ensuring the safety of the student-athletes, and fixing technique and pacing
- Obtained CPR and coaching certifications

AI4SaR Instructional Student Assistant | Cal Poly SLO

Jun. 2024 - Dec. 2024

- Spearheaded research and development of an AI-driven project locating missing persons using public transit
- Developed the Public Transit Search Widget, a predictive map tool indicating likely movements of missing persons
- Employed TensorFlow/Scikit-Learn to build 6 models: Decision Tree, Random Forest, ANN, RNN, LSTM, and BiLSTM
- Implemented Folium and Plotly for data analysis and geospatial mapping

COMPUTER SCIENCE EXPERIENCE

- **LLM Research:** Produced 2 LLM's, using PyTorch and LLM architecture/features
 - Base LLM: 33 thousand parameter neural network using embeddings
 - Mock Llama: 2 million parameter LLM, implementing RMSNorm pre-normalization, SwiGLU activation function, rotary position embeddings, multi-head attention, GPU/Cuda usage
- **Wikipedia Game Solver:** Implemented 4 unsupervised and supervised AI algorithms (BFS, DFS, GBFS, and A*) to solve the Wikipedia Game. Developed a report and analysis on the results
- **Data Science For Cars:** Analysis of cars for college students. Used KNN and KMeans clustering on a Kaggle dataset of used cars. Cleaning, making models of, and graphically analyzing the data
- **Github Portfolio:** Portfolio website built as a simplified mockup of GitHub. Uses Typescript, React, and Vite to deploy a responsive site via GitHub Pages.
- **Tiny File System:** Created a tiny file system (TFS) that can be mounted on a UNIX file, supporting files, defragmentation, timestamps, and read/write permissions